

Hardware Connections (I2C Interface)

The GY-271 module communicates with the Arduino using the I2C protocol. Assuming you are using a standard Arduino board like the Uno or Nano, connect the pins as follows:

- **VCC** on sensor $\rightarrow$ **5V** (or 3.3V) on Arduino
- **GND** on sensor $\rightarrow$ **GND** on Arduino
- **SCL** on sensor $\rightarrow$ **A5** on Arduino Uno/Nano (*For Arduino Mega, use pin 21*)
- **SDA** on sensor $\rightarrow$ **A4** on Arduino Uno/Nano (*For Arduino Mega, use pin 20*)

(Note: If your specific Arduino board has dedicated SDA and SCL pins printed on the header, you can use those instead).

Library Installation

Your code requires the QMC5883LCompass.h library to function. You can easily install this directly within the Arduino IDE.

Steps to install via the Library Manager:

1. Open the Arduino IDE.
2. Go to the top menu and click on **Sketch** $\rightarrow$ **Include Library** $\rightarrow$ **Manage Libraries...** (Alternatively, press Ctrl + Shift + I on Windows/Linux or Cmd + Shift + I on Mac).
3. In the Library Manager window, locate the search bar and type **QMC5883LCompass**.
4. Look for the library specifically named **QMC5883LCompass** (often authored by *MRPrograms* or *MPrograms*).
5. Click the **Install** button next to it.
6. Once the installation is complete, close the Library Manager.

After connecting the pins and installing the library, you can upload the code to your Arduino, open the **Serial Monitor** (set the baud rate to **9600**), and you will start seeing the X, Y, Z, and Heading values.